

Designing Interactive Experiences in the Interplay between Ambient Intelligence and Mixed Reality

Radu-Daniel Vatavu

MintViz Lab, MANSiD Research Center
Ștefan cel Mare University of Suceava
Suceava, Romania
radu.vatavu@usm.ro

ABSTRACT

Our physical environments integrate more and more sensing, processing, and communications technology toward the realization of the vision of Ambient Intelligence (AmI). At the same time, virtual worlds are increasingly more linked to physical spaces by means of applications delivering the vision of Augmented and Mixed Reality (AR/MR). In this context, several opportunities emerge at the intersection of these two visions of computing. The goal of this course is to achieve the realization that AmI and MR are two sides of the same coin in order to deliver a fresh perspective to HCI researchers and practitioners interested in designing interactive experiences beyond the desktop and mobile computing paradigms by employing the concepts, principles, and methods of AmI and MR.

CCS CONCEPTS

• **Human-centered computing** → **Ambient intelligence; Mixed / augmented reality; Ubiquitous and mobile computing systems and tools; HCI theory, concepts and models.**

KEYWORDS

Ambient intelligence, mixed reality, augmented reality, extended reality, interaction design, methods, examples, software tools

ACM Reference Format:

Radu-Daniel Vatavu. 2022. Designing Interactive Experiences in the Interplay between Ambient Intelligence and Mixed Reality. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29-May 5, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3491101.3503763>

1 INTRODUCTION AND CONTEXT

The vision of smart environments [1,6] that are responsive to the presence and actions of users, but also anticipatory and adaptive in their responses, has already entered mainstream with various products and services, such as smart lighting, personal voice assistants, personalized on-demand streaming services, and IoT protocols and standards implemented by smart objects designed for the home. At the same time, the original visions of Augmented and Mixed Reality (AR/MR) [2,14] are now at the fingertips of end users in the form

of applications for mobile devices, smartglasses, and head-mounted displays (HMDs) with a steadily growing consumer market.¹ In this context, designing interactive experiences for smart environments that integrate sensing, processing, and communications technology in the physical surroundings and everyday objects, but also for environments that are hybrid versions of the physical and the virtual has never been more actual and needed. *The unique perspective of this course comes from the interplay between AmI and MR environments, which enables opportunities for the design of new interactive experiences at the intersection of these two areas of scientific investigation and practical development.* The realization that AmI and MR are two sides of the same coin will be helpful for HCI researchers and practitioners that are looking for a new perspective to employ in their work. This course offers this realization by taking the attendees in a journey represented by a critical analysis of the theoretical foundations of AmI and MR accompanied by practical examples of systems, applications, and tools from AmI and MR.

2 DESCRIPTION

The course will introduce attendees to the theoretical foundations and practical aspects of designing interactions for AmI and MR.² Relevant references from the scientific literature will be discussed and software tools demonstrated; see the next subsection for examples. The course will be offered in the form of two sessions of 75 minutes each, pointing attendees to the most relevant aspects of designing interactions in the interplay of AmI and MR by combining expositions and practical activities. The first session addresses theoretical foundations—principles and concepts for AmI and MR—accompanied by examples from the scientific literature. The second session focuses on methods for designing interactions in the interplay between AmI and MR environments, featuring more examples as well as a practical activity. A dedicated web page is available for this course with notes and references for the attendees. Key literature and tools from the course are linked to this page.

2.1 Content

Part 1: Theoretical foundations of AmI and MR

- ① *Introduction.* Setting the context and overview of the course.
- ② *The user experience of Ambient Intelligence environments.* An exposition of the design principles and quality characteristics of

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI '22 Extended Abstracts, April 29-May 5, 2022, New Orleans, LA, USA

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9156-6/22/04.

<https://doi.org/10.1145/3491101.3503763>

¹Forecast augmented (AR) and virtual reality (VR) market size worldwide from 2020 to 2024. Available from <https://www.statista.com/statistics/591181/global-augmented-virtual-reality-market-size>, last accessed December, 2021.

²According to Milgram and Kishino's [14] Reality-Virtuality Continuum, we interpret AR as a subset of MR. As we discuss in this course the interplay between AmI and MR, MR also includes AR.

AmI environments [1,6,7,26], such as integration, personalization, and anticipation, among others, with a focus on the user experience design [11], such as metaphors for AmI. Examples of tools, including SAPIENS [23] for implementing interactions in smart environments involving a diversity of I/O mobile, wearable, and ambient devices, the flower model for ambient media [16], and databases to inform interactions with specific devices [37], will be discussed. *Benefits:* attendees will discover opportunities offered by interactive AmI environments in the context of the original vision of AmI, with many practical references to use in their own work.

④ *The user experience of Mixed Reality environments.* An exposition of the principles and foundations of AR and MR [2–5,13,14,24,25], such as AR as intelligence amplification, AR as a new form of media, AR as an interface and gateway to a 1:1 correspondence between the virtual and the physical world, and AR as a subset of MR. Practical examples will be provided to the attendees, such as AR for television [39], interacting with mid-air virtual content [20], and specific interaction techniques, such as Smart Pockets [33], designed to augment physical reality for various sensory channels. *Benefits:* attendees will discover opportunities offered by interactive AR/MR environments in the context of the original visions of AR/MR.

Part 2: The interplay between AmI and MR

The second part of the course builds on the foundations set in the first session to achieve the realization of the interconnection between AmI and MR and also to illustrate new design opportunities available in the interplay between AmI and MR.

⑤ *Designing interactions for AmI and MR, two faces of the same coin.* An exposition of the similarities of the visions of AmI and MR by means of commonly shared characteristics of systems implementing these visions toward *adaptation*, *personalization*, and *context awareness*, while other characteristics, such as *integration*, are conceptually similar. Examples will be provided of AR/MR systems that can be described from an AmI perspective, such as IllumiRoom [10] vs. AroundTV [29] vs. Ambilight³ as well as of AmI systems that can be described using AR/MR characteristics, such as Hover [17] or “I bet you look good on the wall” [9]. *Benefits:* the realization that AmI and MR intersect significantly and that their intersection can be creatively exploited for designing novel interactive experiences.

⑥ *Group work: ideation session on a given topic.* Members of the audience will form teams (random composition of groups) and apply the realization ⑤ to their own work, previous projects, and research interests about interactive systems. Each group will present briefly their result and highlight the interplay between AmI and MR. *Benefits:* this part of the course will create the opportunity for the attendees to get to know each other, interact, and collaborate by exchanging ideas and apply the findings from the course.

2.2 Audience and Prerequisites

The level of the course is introductory. No previous knowledge about AmI or AR/MR is required. The ideal audience is researchers and practitioners that wish to design new interactive experiences

involving MR and AmI concepts. This audience includes experienced researchers, UX and interaction design professionals that use the course to get acquainted to AmI and MR terminology, concepts, principles, and methods, but also HCI practitioners with an interest in exploring interactions beyond desktop and mobile computing scenarios, and PhD students that wish to expand their range of knowledge and expertise to gain new perspectives to apply in their own research. The course is planned to be delivered online in order to foster broad participation. To this end, the Google Cloud Platform will be used to host resources, notes, and communication channels between the instructors and attendees.

3 INSTRUCTOR

Radu-Daniel Vatavu, Ph.D., works in the areas of HCI, AmI, AR/MR, Accessible Computing, and Entertainment Computing with focus on designing novel interactions informed by understanding users’ needs, preferences, and behavior with interactive computer systems. He directs the Machine Intelligence and Information Visualization Lab (MintViz),⁴ an interdisciplinary research laboratory within the MANSiD Research Center at the Ștefan cel Mare University of Suceava. The goal of the MintViz lab is design, development, and evaluation of new artificial intelligence and information visualization technology to support innovative, rich interaction between humans, computers, and environments to advance HCI knowledge. Examples of the instructor’s prior research relevant for this course are interaction techniques and tools for AmI application scenarios [8,12,16,20,22,23,28,33,38] and AR/MR [15,18,19,21,39]. A full list of publications is available from his web page.

Teaching Experience. Radu-Daniel Vatavu is a Professor of Computer Science at the Ștefan cel Mare University of Suceava, where he has been teaching courses in Ambient Intelligence, Augmented Reality, Natural Human Computer Interaction, and Algorithms Design since 2008 for undergraduate and graduate students. He has also delivered courses at CHI’18 [27], CHI’17 [32], CHI’16 [31], TVX’15 [30], and other events. He has participated in the design and development of many software tools that were released publicly under an open license, such as the \$P [34] and \$Q [36] gesture recognizers, AGATe [40], GHoST [35], SAPIENS [20], Euphoria [22], GestuRING [37], and others, of which a selection will be featured in this course.

ACKNOWLEDGMENTS

This work was supported by a grant of the Romanian Ministry of Education and Research, CNCS-UEFISCDI, project number PN-III-P4-ID-PCE-2020-0434 (PCE29/2021), within PNCDI III.

REFERENCES

- [1] Emile H.L. Aarts and Frits Grotenhuis. 2011. Ambient Intelligence 2.0: Towards Synergetic Prosperity. *Journal of Ambient Intelligence and Smart Environments* 3, 1 (2011), 3–11. <http://dx.doi.org/10.3233/AIS-2011-0090>
- [2] Ronald T. Azuma. 1997. A Survey of Augmented Reality. *Presence: Teleoper. Virtual Environ.* 6, 4 (1997), 355–385. <https://doi.org/10.1162/pres.1997.6.4.355>
- [3] Ronald T. Azuma. 2016. The Most Important Challenge Facing Augmented Reality. *Presence: Teleoper. Virtual Environ.* 25, 3 (Dec. 2016), 234–238. https://doi.org/10.1162/PRES_a_00264

³<https://www.philips.com/c-w/country-selectorpage/tv/tv-ambilight.html>

⁴<http://www.eed.usv.ro/mintviz>

- [4] Ronald T. Azuma. 2019. The Road to Ubiquitous Consumer Augmented Reality Systems. *Human Behavior and Emerging Technologies* 1, 1 (2019), 26–32. <https://doi.org/10.1002/hbe2.113>
- [5] Mark Billinghurst, Adrian Clark, and Gun Lee. 2015. A Survey of Augmented Reality. *Found. Trends Hum.-Comput. Interact.* 8, 2–3 (March 2015), 73–272. <https://doi.org/10.1561/11000000049>
- [6] Diane J. Cook, Juan C. Augusto, and Vikramaditya R. Jakkula. 2009. Ambient intelligence: Technologies, Applications, and Opportunities. *Pervasive and Mobile Computing* 5, 4 (2009), 277–298. <https://doi.org/10.1016/j.pmcj.2009.04.001>
- [7] Rob Dunne, Tim Morris, and Simon Harper. 2021. A Survey of Ambient Intelligence. *ACM Comput. Surv.* 54, 4, Article 73 (May 2021), 27 pages. <https://doi.org/10.1145/3447242>
- [8] Bogdan-Florin Gheran, Jean Vanderdonckt, and Radu-Daniel Vatavu. 2018. Gestures for Smart Rings: Empirical Results, Insights, and Design Implications. In *Proceedings of the 2018 Designing Interactive Systems Conference (DIS '18)*. ACM, New York, NY, USA, 623–635. <https://doi.org/10.1145/3196709.3196741>
- [9] Vermeulen J., Slenders J., Luyten K., and Coninx K. 2009. I Bet You Look Good on the Wall: Making the Invisible Computer Visible. In *Proceedings of Aml 2009. Lecture Notes in Computer Science*, vol. 5859. Springer, Berlin, Heidelberg, 196–205. https://doi.org/10.1007/978-3-642-05408-2_24
- [10] Brett R. Jones, Hrvoje Benko, Eyal Ofek, and Andrew D. Wilson. 2013. IllumiRoom: Peripheral Projected Illusions for Interactive Experiences. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '13)*. ACM, New York, NY, USA, 869–878. <https://doi.org/10.1145/2470654.2466112>
- [11] Mike Kuniavsky. 2010. *Smart Things: Ubiquitous Computing User Experience Design*. Morgan Kaufmann, Burlington, MA, USA. <https://doi.org/10.1016/C2009-0-20057-2>
- [12] Jean-Yves Lionel Lawson, Jean Vanderdonckt, and Radu-Daniel Vatavu. 2018. Mass-Computer Interaction for Thousands of Users and Beyond. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems (CHI EA '18)*. ACM, New York, NY, USA, 1–6. <https://doi.org/10.1145/3170427.3188465>
- [13] Steve Mann, Tom Furness, Yu Yuan, Jay Iorio, and Zixin Wang. 2018. All Reality: Virtual, Augmented, Mixed (X), Mediated (X.Y), and Multimeditated Reality. [arXiv:1804.08386 \[cs.HC\]](https://arxiv.org/abs/1804.08386) <https://arxiv.org/abs/1804.08386>
- [14] Paul Milgram and Fumio Kishino. 1994. A Taxonomy of Mixed Reality Visual Displays. *IEICE Transactions on Information and Systems* E77-D, 12 (December 1994), 1321–1329. https://search.ieice.org/bin/summary.php?id=e77-d_12_1321
- [15] Cristian Pamparau and Radu-Daniel Vatavu. 2021. FlexiSee: Flexible Configuration, Customization, and Control of Mediated and Augmented Vision for Users of Smart Eyewear Devices. *Multimedia Tools and Applications* 80 (2021), 30943–30968. <https://doi.org/10.1007/s11042-020-10164-5>
- [16] Bogdan Pogorelec, Artur Lugmayr, Bjorn Stockleben, Radu-Daniel Vatavu, Nina Tahmasebi, Estefania Serral, Emilija Stojmenova, Bojan Imperl, Thomas Risse, Gideon Zenz, and Matjaz Gams. 2013. Ambient Bloom: New Business, Content, Design and Models to Increase the Semantic Ambient Media Experience. *Multimedia Tools and Applications* 66, 1 (2013), 7–32. <http://dx.doi.org/10.1007/s11042-012-1228-4>
- [17] Irina Popovici, Ovidiu-Andrei Schipor, and Radu-Daniel Vatavu. 2019. Hover: Exploring cognitive maps and mid-air pointing for television control. *International Journal of Human-Computer Studies* 129 (2019), 95–107. <https://doi.org/10.1016/j.ijhcs.2019.03.012>
- [18] Irina Popovici and Radu-Daniel Vatavu. 2019. Understanding Users' Preferences for Augmented Reality Television. In *Proceedings of the 18th IEEE International Symposium on Mixed and Augmented Reality (ISMAR '19)*. IEEE, Washington, DC, USA, 269–278. <https://doi.org/10.1109/ISMAR.2019.00024>
- [19] Irina Popovici, Radu-Daniel Vatavu, Pu Feng, and Wenjun Wu. 2021. AR-TV and AR-Diànshi: Cultural Differences in Users' Preferences for Augmented Reality Television. In *ACM International Conference on Interactive Media Experiences (IMX '21)*. ACM, New York, NY, USA, 50–60. <https://doi.org/10.1145/3452918.3458801>
- [20] Ovidiu-Andrei Schipor and Radu-Daniel Vatavu. 2018. Invisible, Inaudible, and Impalpable: Users' Preferences and Memory Performance for Digital Content in Thin Air. *IEEE Pervasive Computing* 17, 4 (2018), 76–85. <https://doi.org/10.1109/MPRV.2018.2873856>
- [21] Ovidiu-Andrei Schipor and Radu-Daniel Vatavu. 2021. Empirical Results for High-definition Video and Augmented Reality Content Delivery in Hyper-connected Cars. *Interacting with Computers* 33, 1 (2021), 3–16. <https://doi.org/10.1093/iwcomp/iwaa025>
- [22] Ovidiu-Andrei Schipor, Radu-Daniel Vatavu, and Jean Vanderdonckt. 2019. Euphoria: A Scalable, Event-Driven Architecture for Designing Interactions Across Heterogeneous Devices in Smart Environments. *Information and Software Technology* 109 (2019), 43–59. <https://doi.org/10.1016/j.infsof.2019.01.006>
- [23] Ovidiu-Andrei Schipor, Radu-Daniel Vatavu, and Wenjun Wu. 2019. SAPIENS: Towards Software Architecture to Support Peripheral Interaction in Smart Environments. *Proc. ACM Hum.-Comput. Interact.* 3, EICS, Article 11 (June 2019), 24 pages. <https://doi.org/10.1145/3331153>
- [24] Richard Skarbez, Missie Smith, and Mary C. Whitton. 2021. Revisiting Milgram and Kishino's Reality-Virtuality Continuum. *Frontiers in Virtual Reality* 2 (2021), 27. <https://doi.org/10.3389/frvir.2021.647997>
- [25] Maximilian Speicher, Brian D. Hall, and Michael Nebeling. 2019. What is Mixed Reality?. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 1–15. <https://doi.org/10.1145/3290605.3300767>
- [26] Norbert Streitz, Dimitris Charitos, Maurits Kaptein, and Marc Böhlen. 2019. Grand challenges for ambient intelligence and implications for design contexts and smart societies. *Journal of Ambient Intelligence and Smart Environments* 11 (2019), 87–107. <http://dx.doi.org/10.3233/AIS-180507>
- [27] Jean Vanderdonckt and Radu-Daniel Vatavu. 2018. Designing, Engineering, and Evaluating Gesture User Interfaces. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems (CHI EA '18)*. ACM, New York, NY, USA, 1–4. <https://doi.org/10.1145/3170427.3170648>
- [28] Radu-Daniel Vatavu. 2013. On Designing Interactivity Awareness for Ambient Displays. *Multimedia Tools and Applications* 66, 1 (2013), 59–80. <http://dx.doi.org/10.1007/s11042-012-1140-y>
- [29] Radu-Daniel Vatavu. 2013. There's a World Outside Your TV: Exploring Interactions beyond the Physical TV Screen. In *Proceedings of the 11th European Conference on Interactive TV and Video (EuroITV '13)*. ACM, New York, NY, USA, 143–152. <https://doi.org/10.1145/2465958.2465972>
- [30] Radu-Daniel Vatavu. 2015. Gesture Interfaces, Ambient Intelligence, and Augmented Reality for the Interactive TV. In *Proceedings of the ACM International Conference on Interactive Experiences for TV and Online Video (TVX '15)*. ACM, New York, NY, USA, 197–198. <https://doi.org/10.1145/2745197.2745698>
- [31] Radu-Daniel Vatavu. 2016. Tools for Designing for Home Entertainment: Gesture Interfaces, Augmented Reality, and Smart Spaces. In *Proceedings of the CHI Conference Extended Abstracts on Human Factors in Computing Systems (CHI EA '16)*. ACM, New York, NY, USA, 1003–1006. <https://doi.org/10.1145/2851581.2856676>
- [32] Radu-Daniel Vatavu. 2017. Fundamentals of Gesture Production, Recognition, and Analysis. In *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems (CHI EA '17)*. ACM, New York, NY, USA, 1174–1177. <https://doi.org/10.1145/3027063.3027106>
- [33] Radu-Daniel Vatavu. 2017. Smart-Pockets: Body-Deictic Gestures for Fast Access to Personal Data during Ambient Interactions. *International Journal of Human-Computer Studies* 103 (2017), 1–21. <https://doi.org/10.1016/j.ijhcs.2017.01.005>
- [34] Radu-Daniel Vatavu, Lisa Anthony, and Jacob O. Wobbrock. 2012. Gestures as Point Clouds: A \$P Recognizer for User Interface Prototypes. In *Proceedings of the 14th ACM International Conference on Multimodal Interaction (ICMI '12)*. ACM, New York, NY, USA, 273–280. <https://doi.org/10.1145/2388676.2388732>
- [35] Radu-Daniel Vatavu, Lisa Anthony, and Jacob O. Wobbrock. 2014. Gesture Heatmaps: Understanding Gesture Performance with Colorful Visualizations. In *Proceedings of the 16th International Conference on Multimodal Interaction (ICMI '14)*. ACM, New York, NY, USA, 172–179. <https://doi.org/10.1145/2663204.2663256>
- [36] Radu-Daniel Vatavu, Lisa Anthony, and Jacob O. Wobbrock. 2018. \$Q: A Super-Quick, Articulation-Invariant Stroke-Gesture Recognizer for Low-Resource Devices. In *Proceedings of the 20th International Conference on Human-Computer Interaction with Mobile Devices and Services (MobileHCI '18)*. ACM, New York, NY, USA, Article 23, 12 pages. <https://doi.org/10.1145/3229434.3229465>
- [37] Radu-Daniel Vatavu and Laura-Bianca Bilius. 2021. GestuRING: A Web-Based Tool for Designing Gesture Input with Rings, Ring-Like, and Ring-Ready Devices. In *Proceedings of the 34th Annual ACM Symposium on User Interface Software and Technology (UIST '21)*. ACM, New York, NY, USA, 710–723. <https://doi.org/10.1145/3472749.3474780>
- [38] Radu-Daniel Vatavu, Catalin Marian Chera, and Wei-Tek Tsai. 2012. Gesture Profile for Web Services: An Event-driven Architecture to Support Gestural Interfaces for Smart Environments. In *Proceedings of the International Joint Conference on Ambient Intelligence. Lecture Notes in Computer Science*, vol. 7683 (AmI'12). Paternò F., de Ruyter B., Markopoulos P., Santoro C., van Loenen E., and Luyten K. (Eds.). Springer, Berlin, Heidelberg, 161–176. http://dx.doi.org/10.1007/978-3-642-34898-3_11
- [39] Radu-Daniel Vatavu, Pejman Saeghe, Teresa Chambel, Vinoba Vinayagamoorthy, and Marian F Ursu. 2020. Conceptualizing Augmented Reality Television for the Living Room. In *Proceedings of the ACM International Conference on Interactive Media Experiences (IMX '20)*. ACM, New York, NY, USA, 1–12. <https://doi.org/10.1145/3391614.3393660>
- [40] Radu-Daniel Vatavu and Jacob O. Wobbrock. 2016. Between-Subjects Elicitation Studies: Formalization and Tool Support. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems (CHI '16)*. ACM, New York, NY, USA, 3390–3402. <https://doi.org/10.1145/2858036.2858228>